Statistical learning theory

Weekend 1, Friday late afternoon

Carlos Cotrini

Friday 4 September 2026

1

Section 1

Empirical risk minimisation

Vladimir Vapnik

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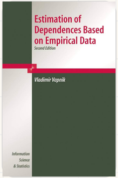

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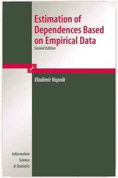

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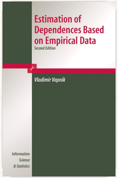

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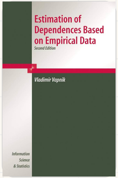

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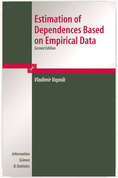

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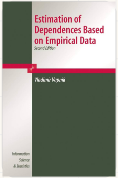

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- **Estimating a probability.** The score is how surprised the model is by what actually happened.
- Three problems that had lived in different books. The score says which one you are doing, and what happens to the score afterwards never changes.

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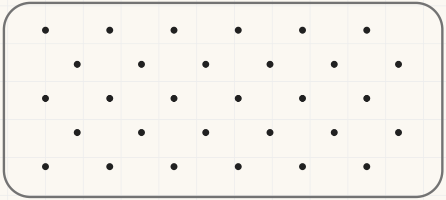
The number we want and the number we have

$\mathcal{M}$ is a candidate model: any rule that turns a row of data into an estimate.

Every case the world will hand us



$R(\mathcal{M})$, the average miss over all of them

Nobody can compute this number

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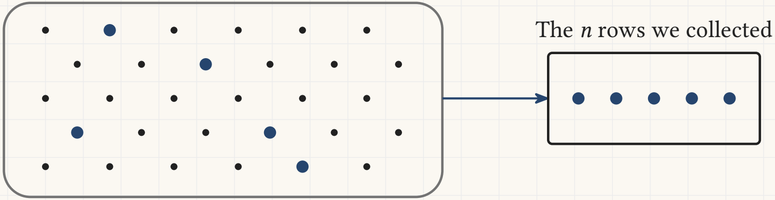

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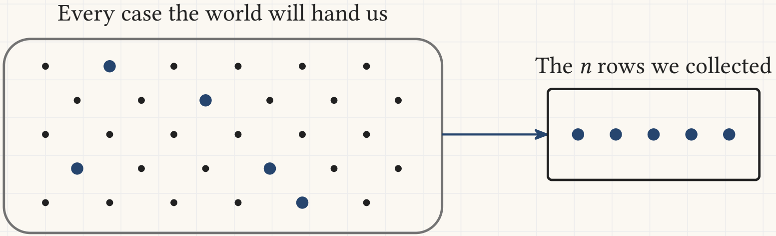

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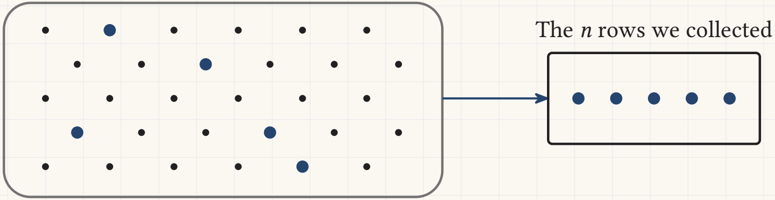

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Empirical risk minimisation: choose the $\mathcal{M}$ that makes $\bar{l}$ small, and take that as the answer to a question about R.

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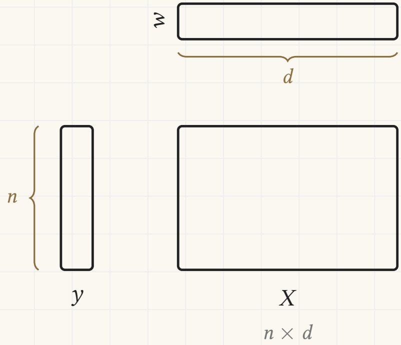
Section 2

The loop we have been running

Linear regression with gradient descent

Two o'clock, in one picture. Each box is a piece of data, and its shape is the shape of the data.



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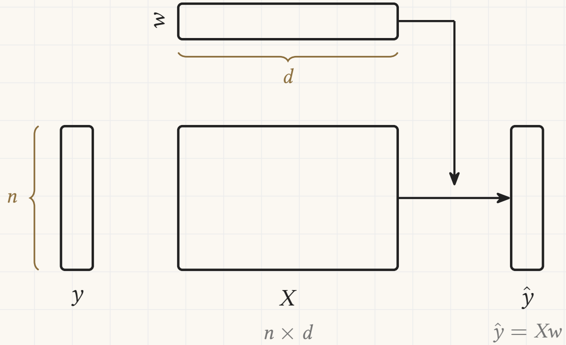

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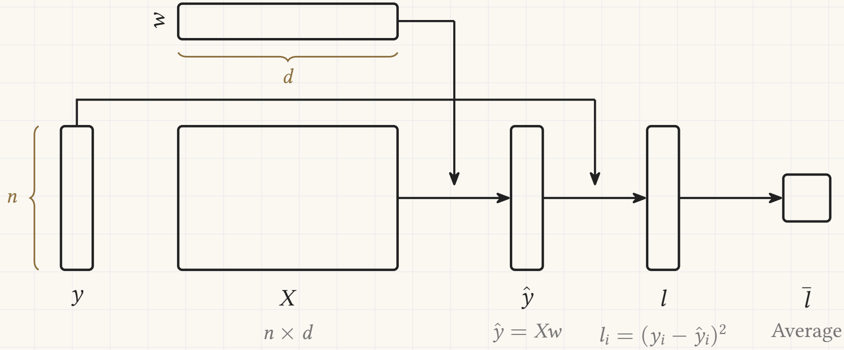

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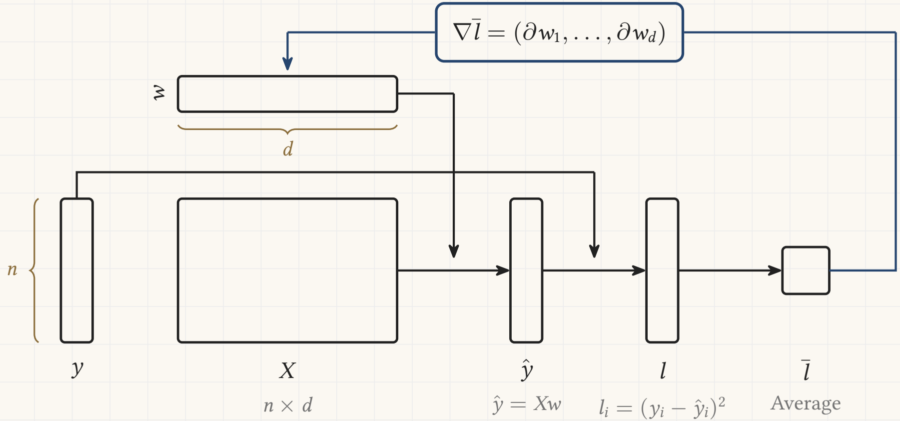
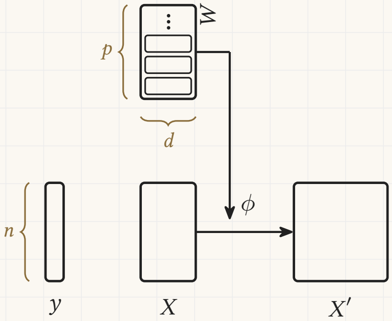


A neural network



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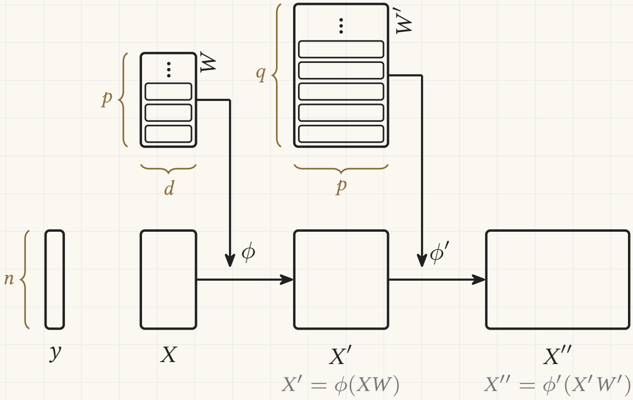

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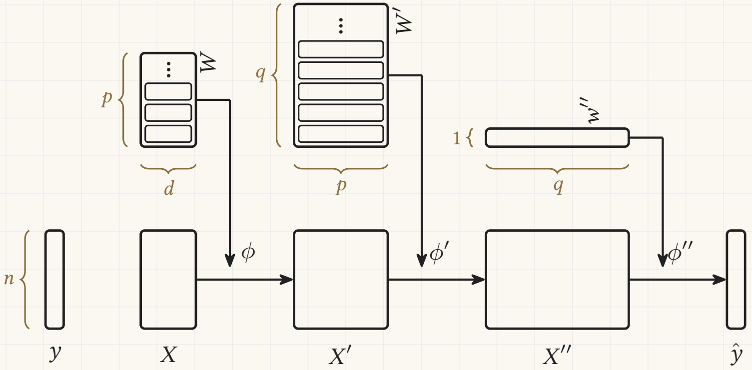

$X' = \phi(XW)$ $X'' = \phi'(X'W')$ $\hat{y} = \phi''(X''w'')$

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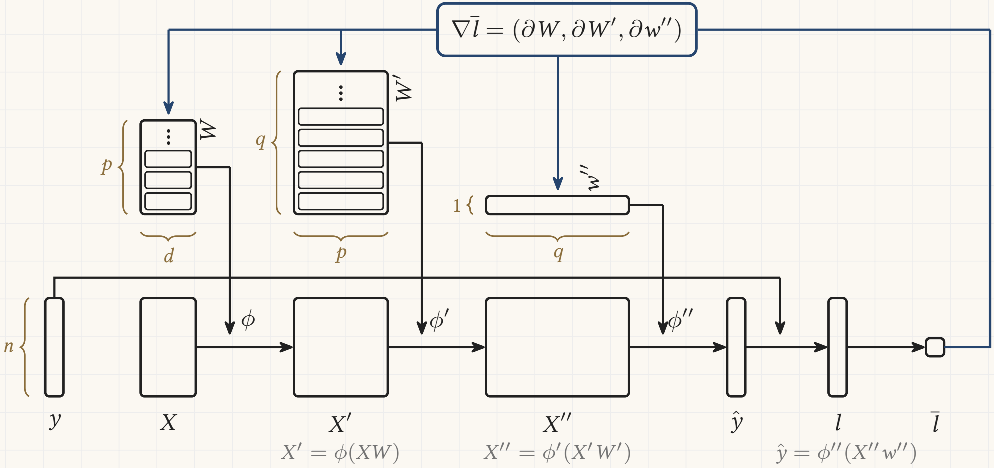
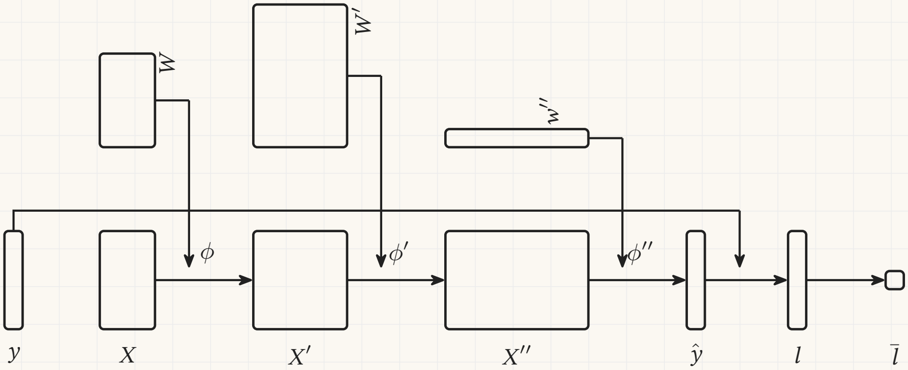


Everything between the data and the estimate

The three weight blocks and the two tables between them are the model. Give the whole of it one name.



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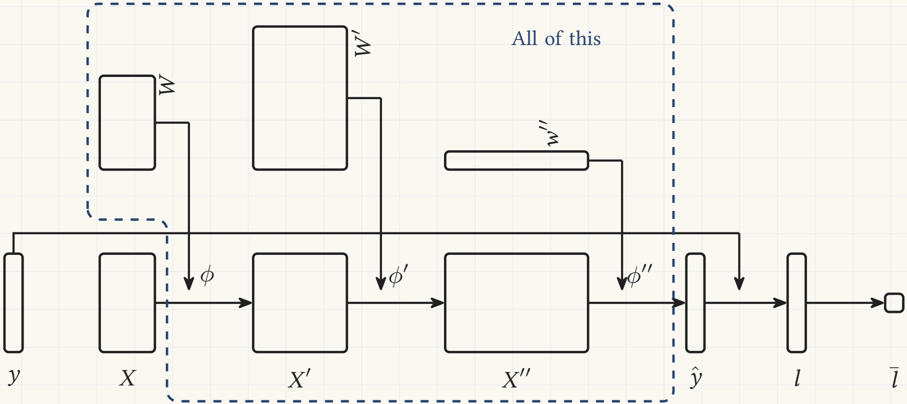

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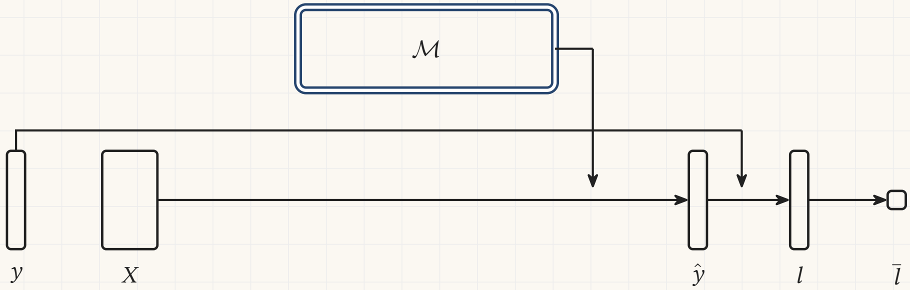

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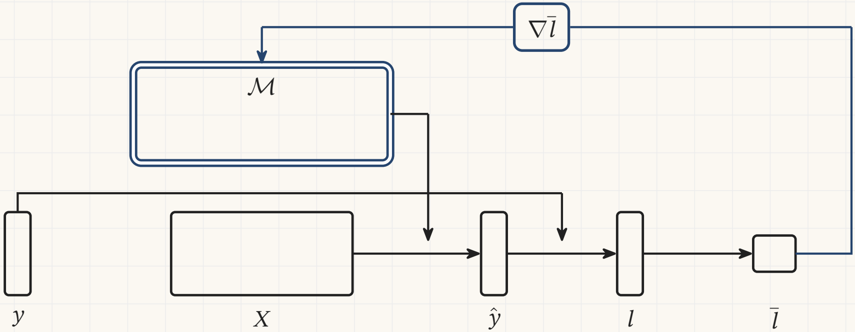
Section 3

The same loop, elsewhere

The loop, with the model as one box

The same picture, with $\mathcal{M}$ where the network was.

The five flats are the ones from this morning, and the network is the one from this afternoon.



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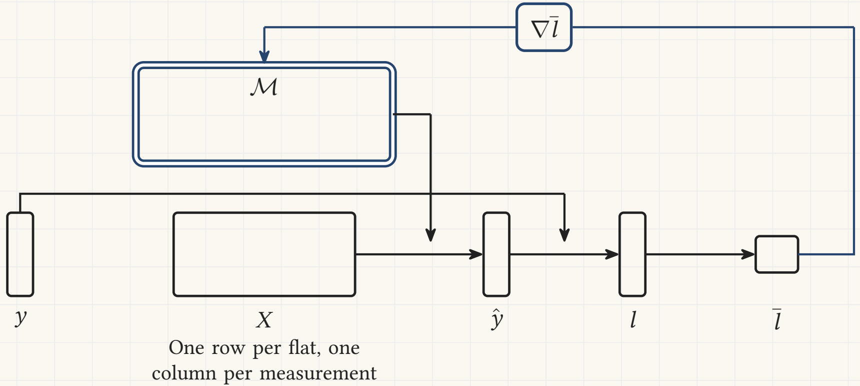

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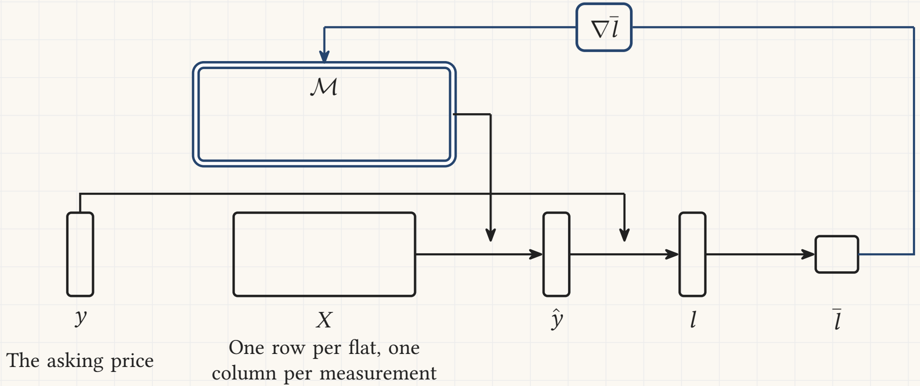

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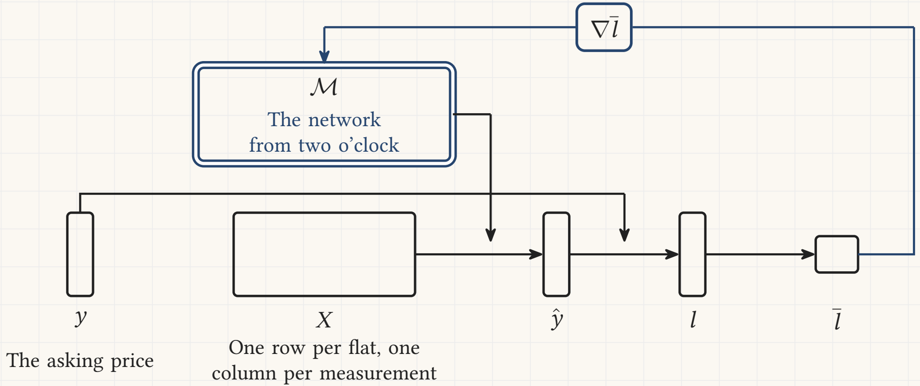

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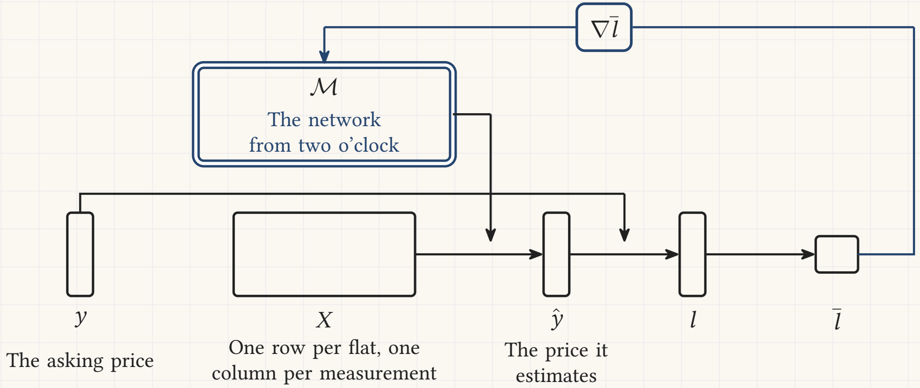
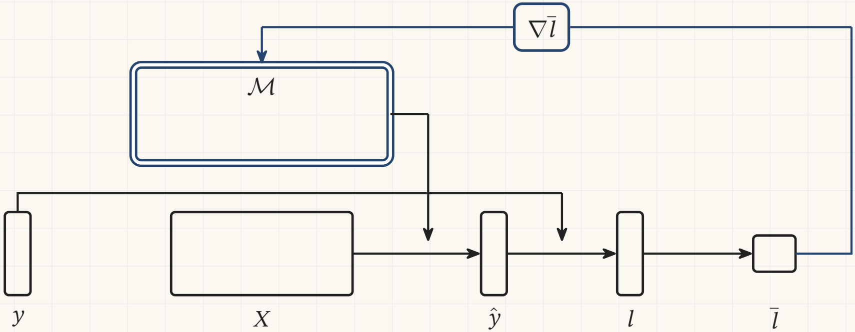


Large language models

A row is text with its next token cut off. Every position of every document gives one.

The box holds a transformer. Vaswani and others, *Attention is all you need*, 2017.



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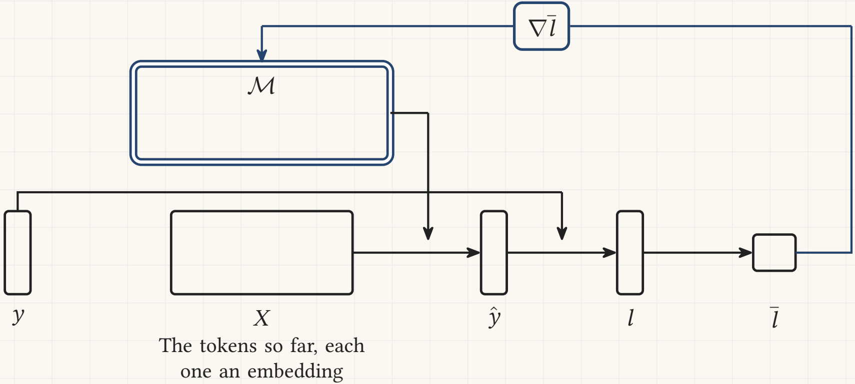

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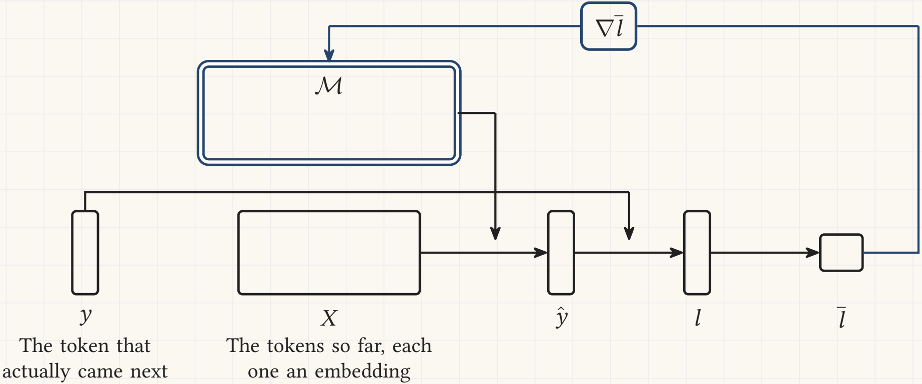

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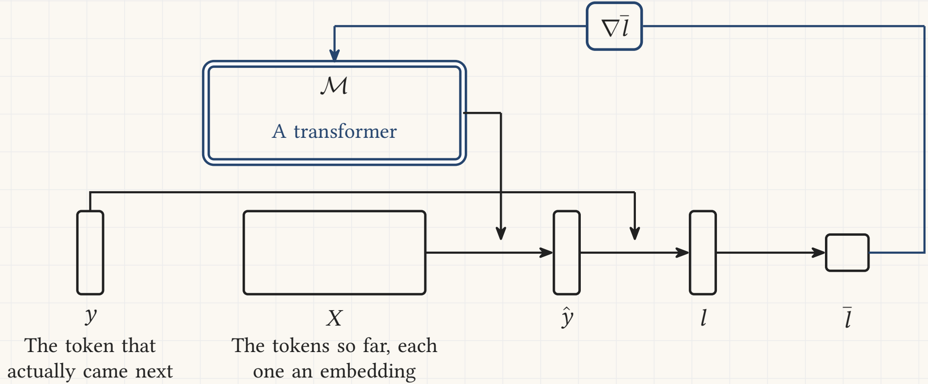

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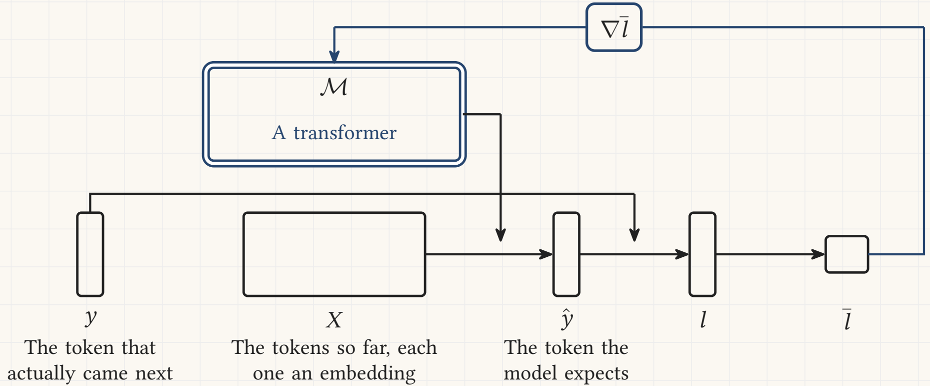
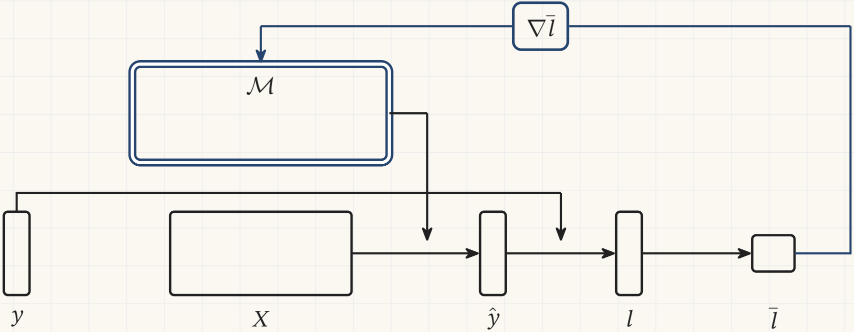


Reasoning models

Start from a trained language model and keep going, on problems a machine can mark.

The verdict is a marker rather than a subtraction. DeepSeek-R1, *Nature* 645, 633 (2025).



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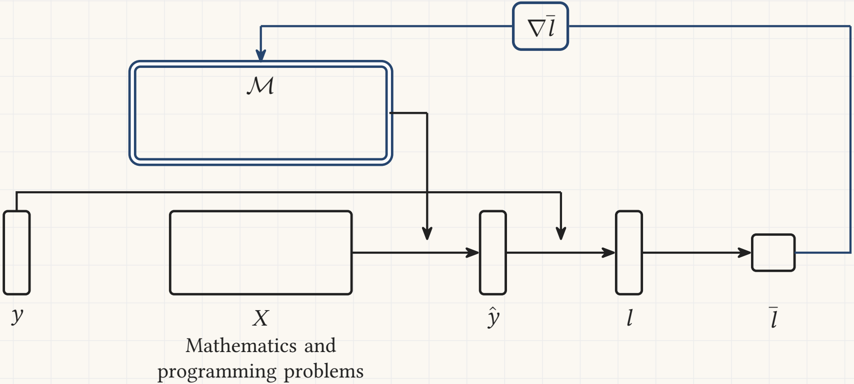

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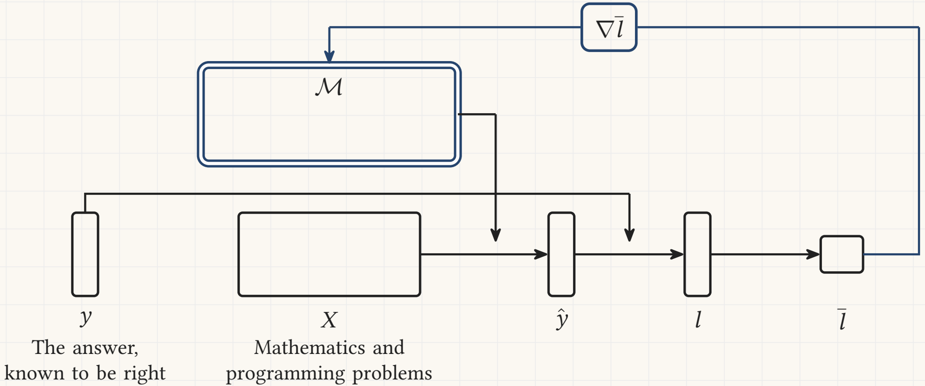

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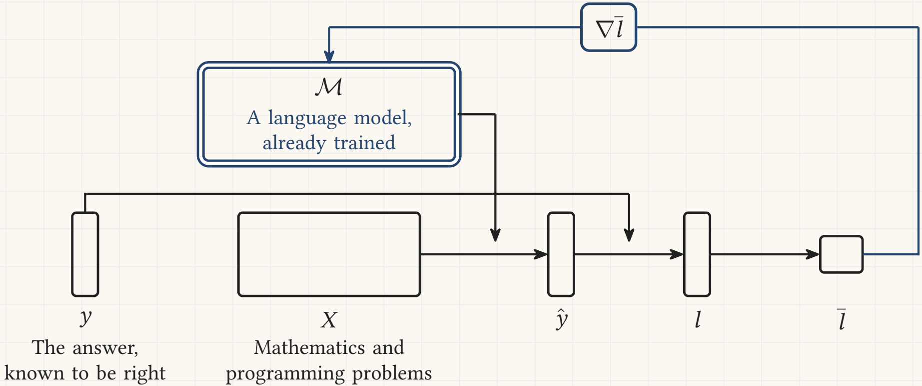

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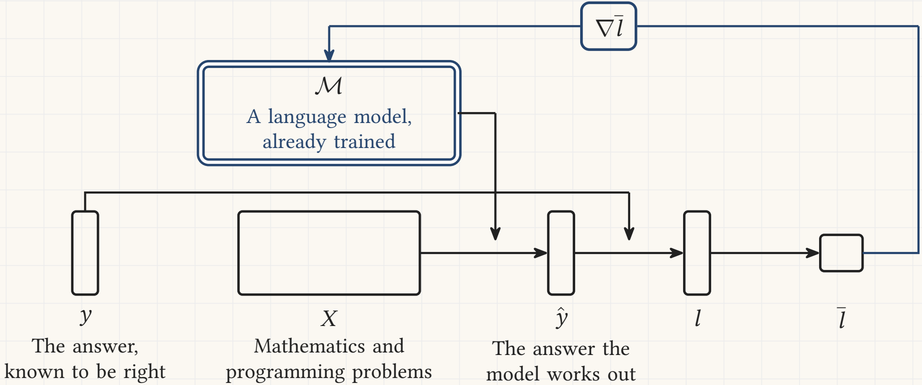

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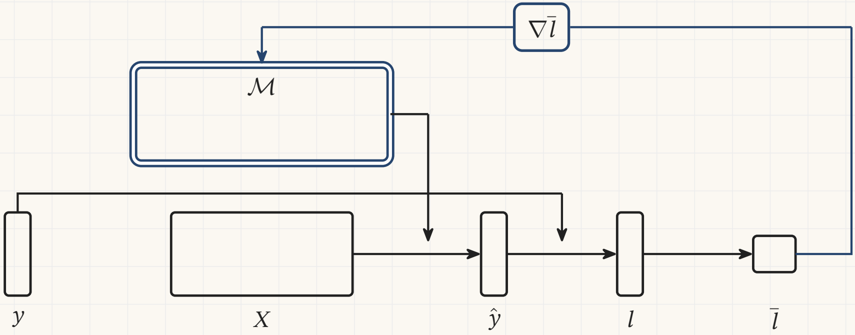

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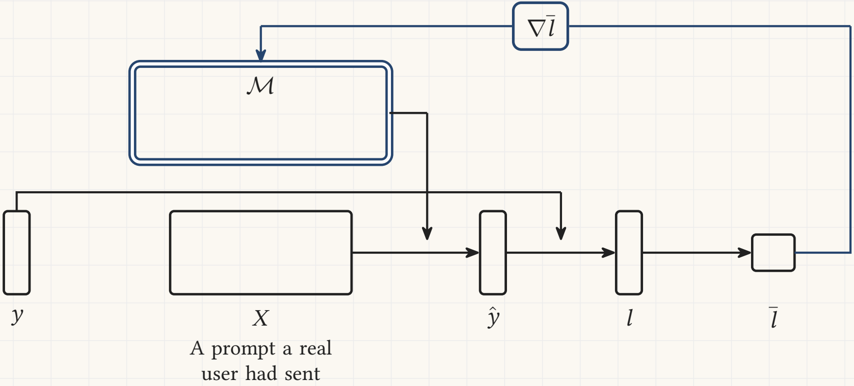

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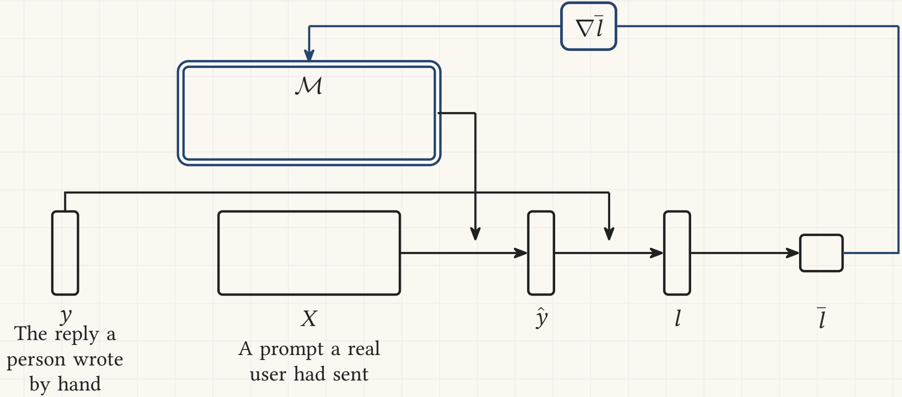

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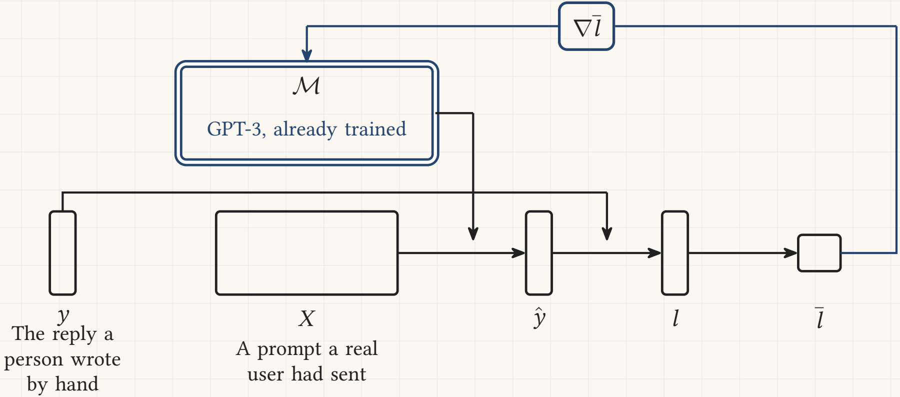

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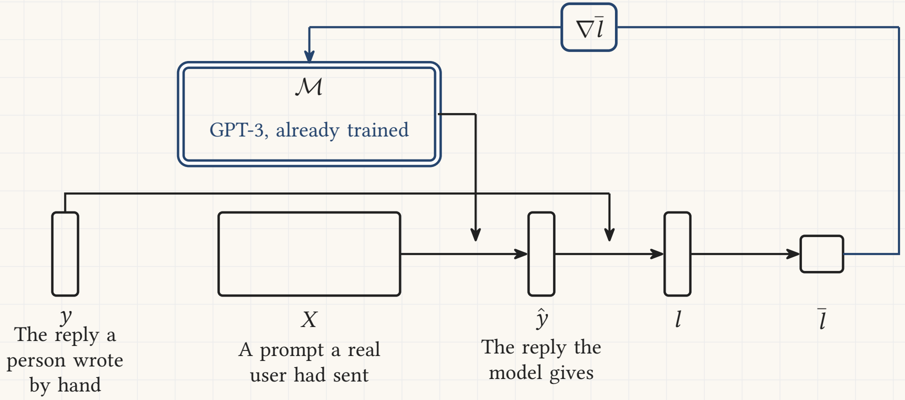

Image generation

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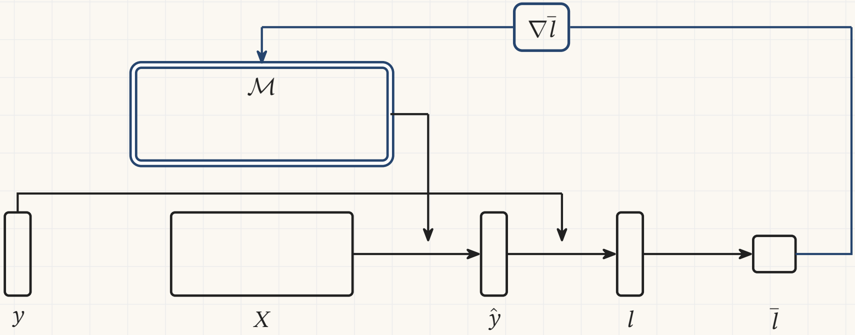
The loss here really is the squared miss. Ho, Jain and Abbeel, 2020; Rombach and others, 2022.



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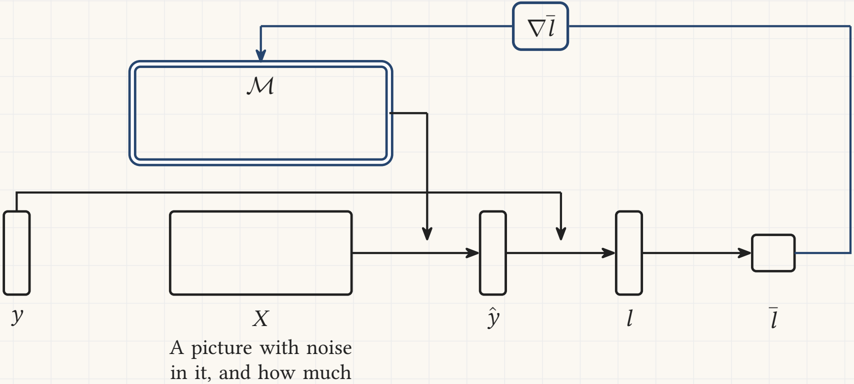

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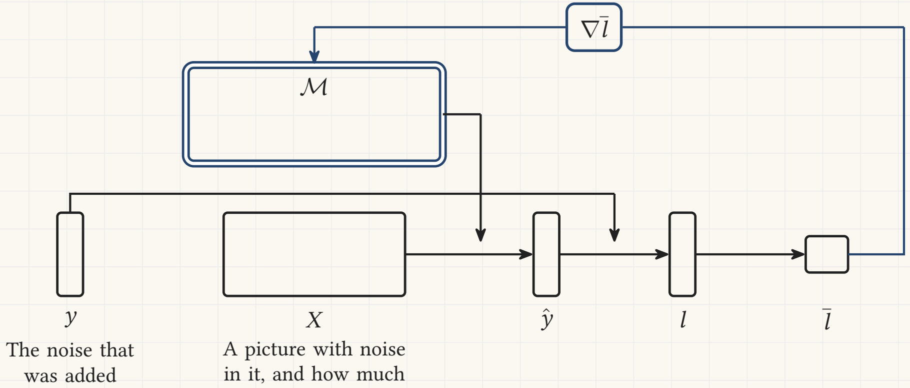

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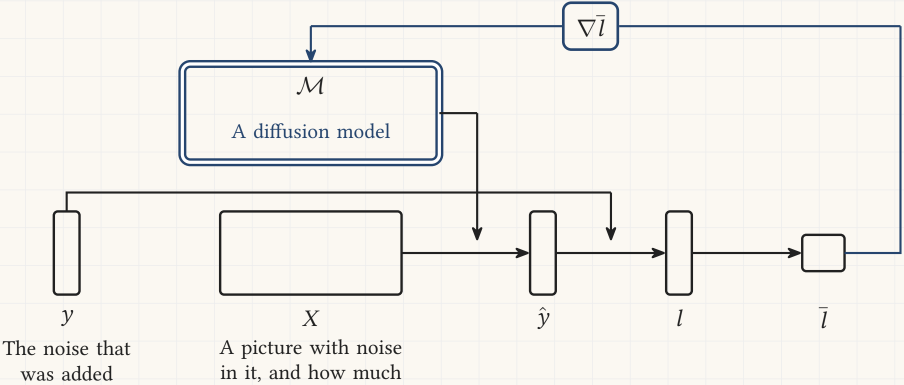

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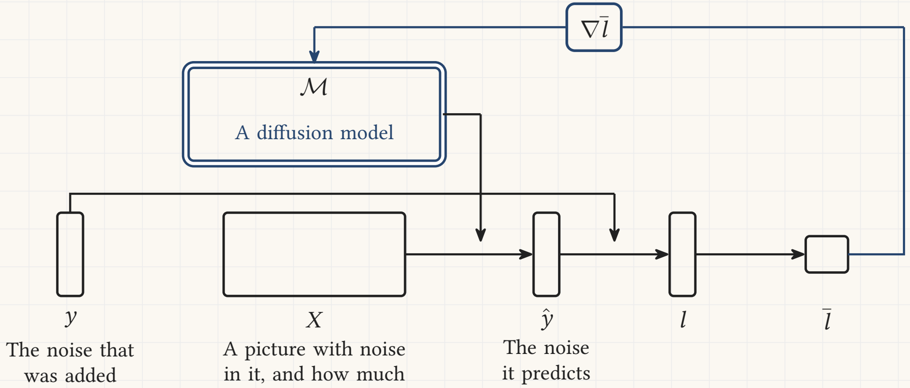
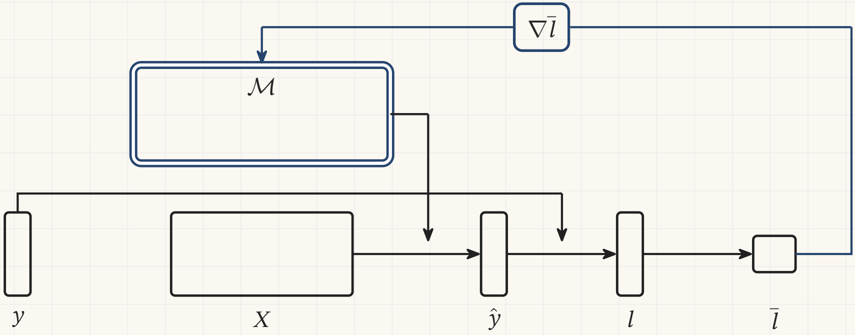


Transcription

Every recording that already carries a transcript is a row. There are 680 000 hours.

Whisper. Radford and others, ICML 2023.



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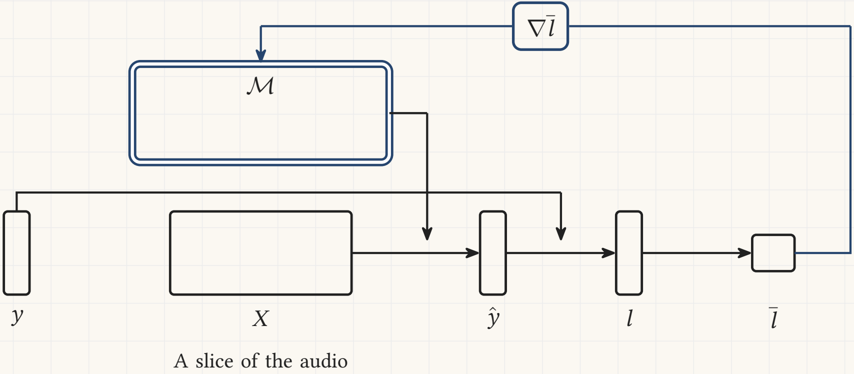

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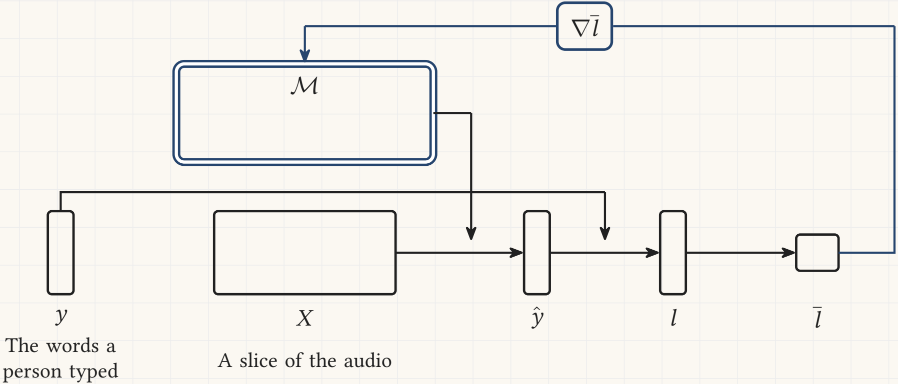

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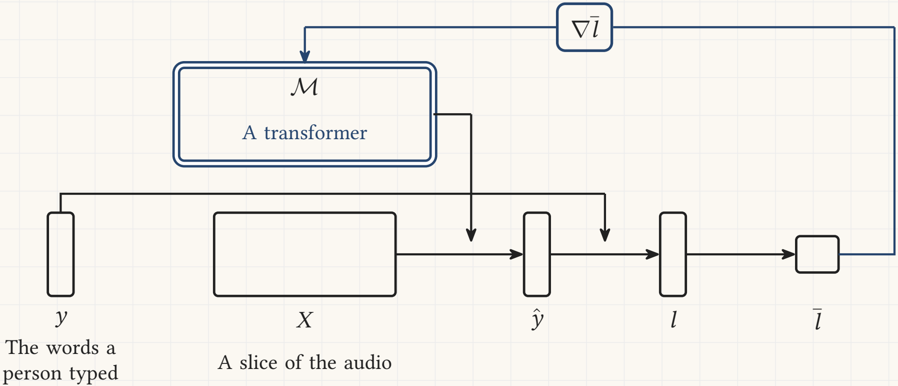

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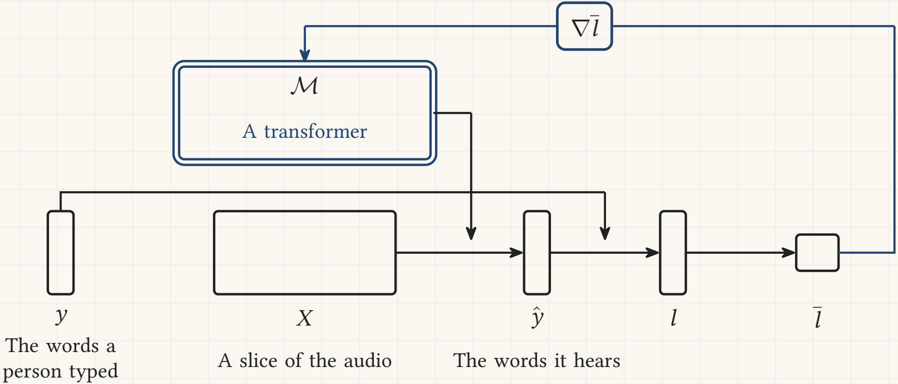
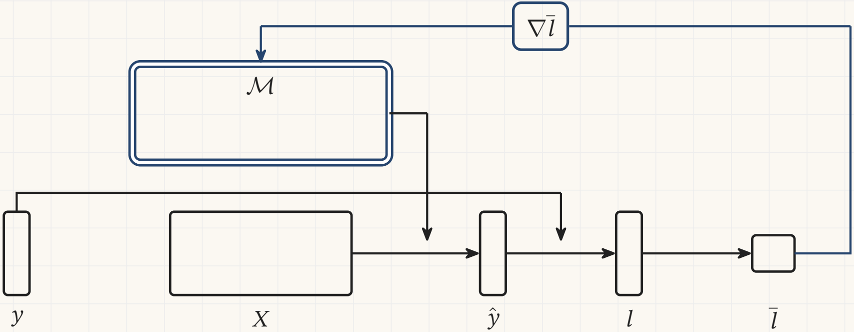


Ranking and recommendation

Every impression is a row: what was shown, to whom, and whether they took it.

The system that picks the next video. Covington, Adams and Sargin, RecSys 2016.



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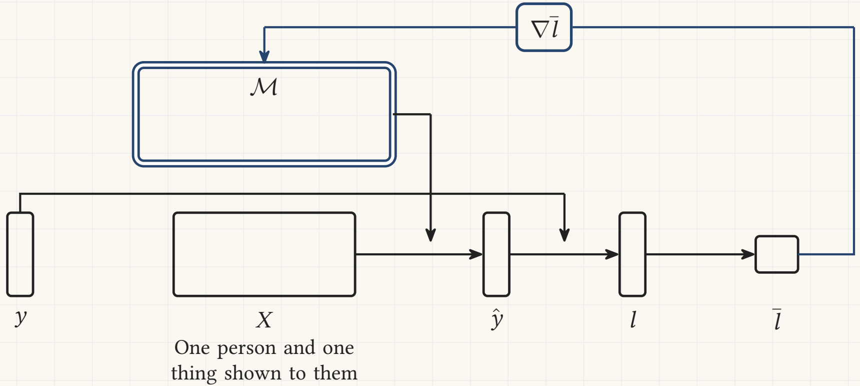

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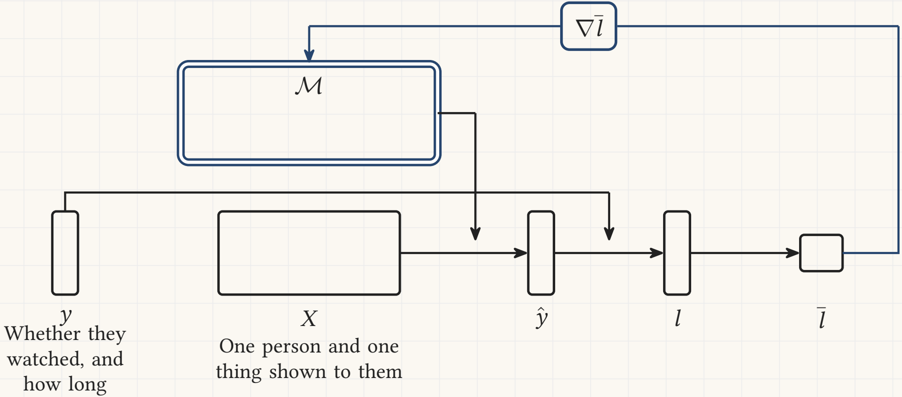

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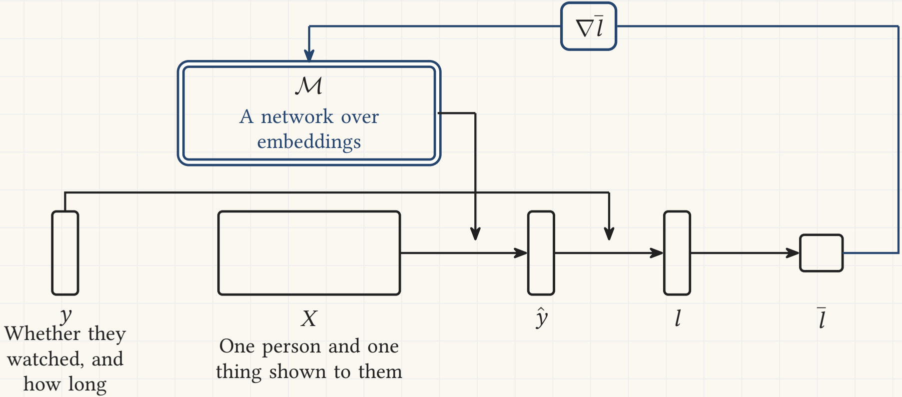

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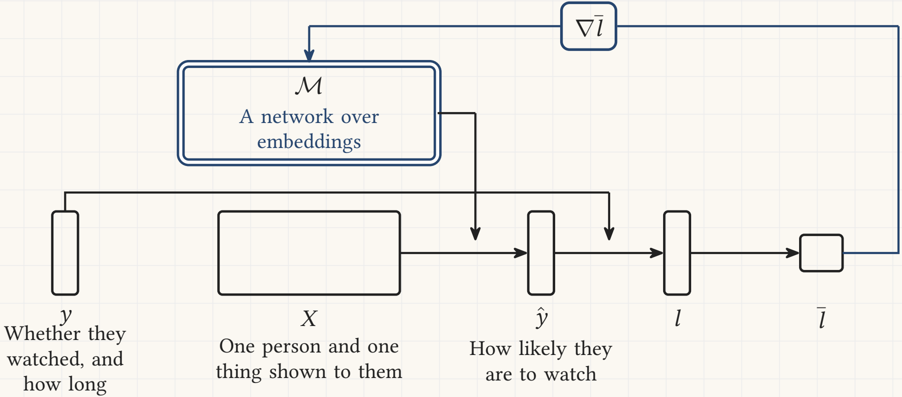
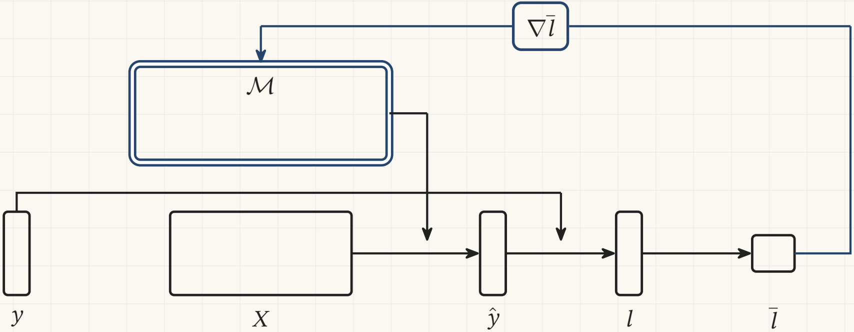


Fraud and credit decisions

Every settled transaction is a row, and its label arrives weeks later with the chargeback.

Dal Pozzolo and others, 2018, with Worldline. The tree ensemble is fitted to the gradient too.



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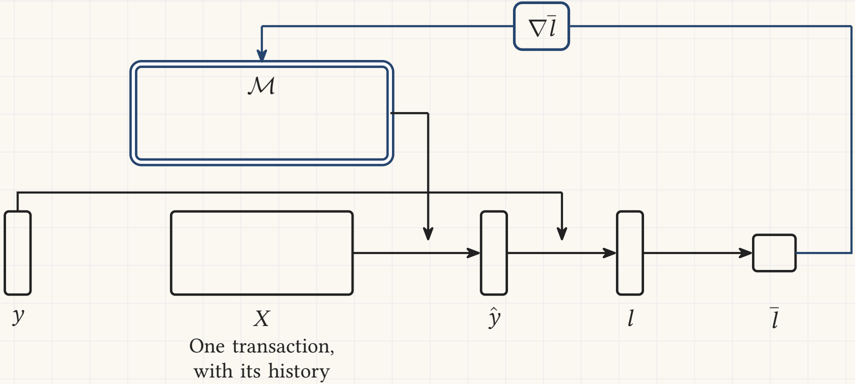

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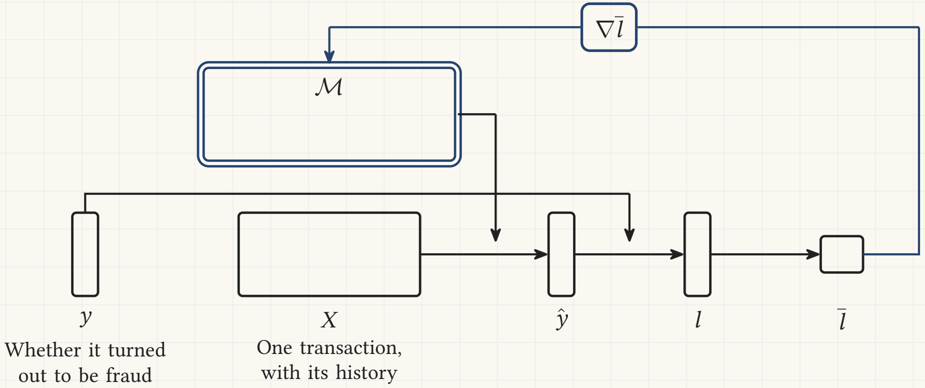

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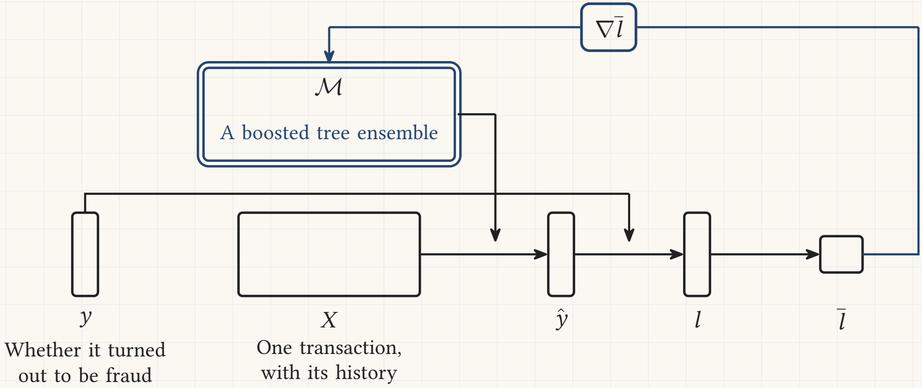

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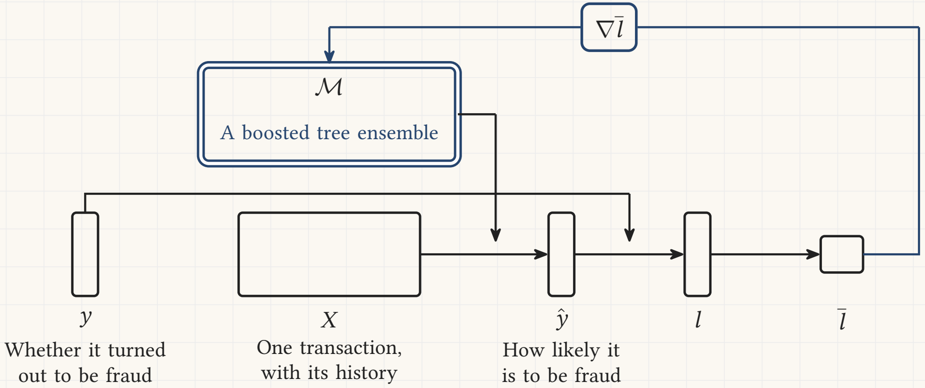
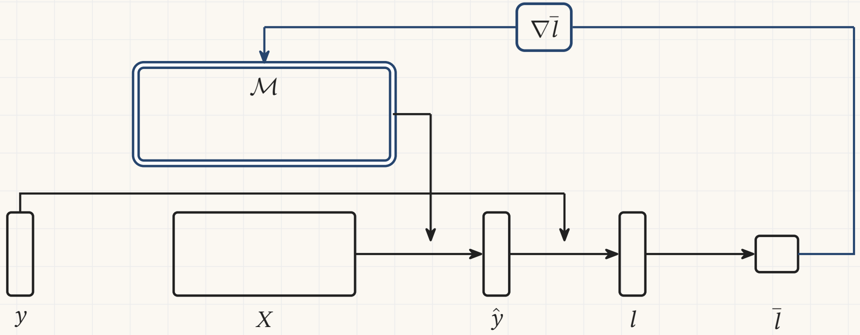


Screening for diabetic eye disease

In 2018 a regulator cleared a machine to read these photographs with no doctor behind it.

128 175 photographs, graded 3 to 7 times by 54 ophthalmologists. Gulshan and others, *JAMA*, 2016.



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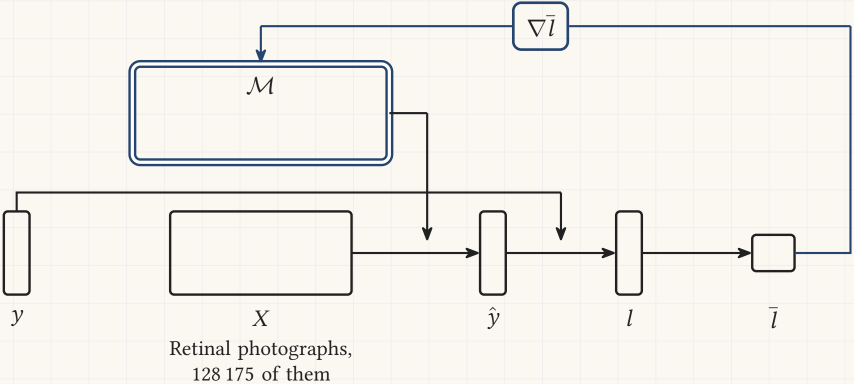

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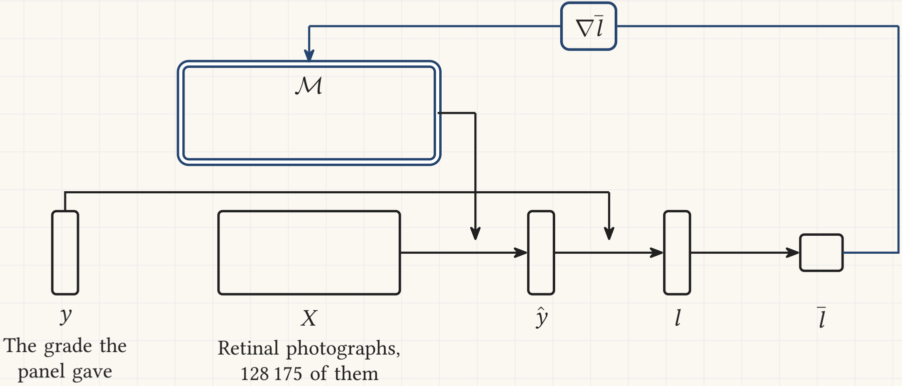

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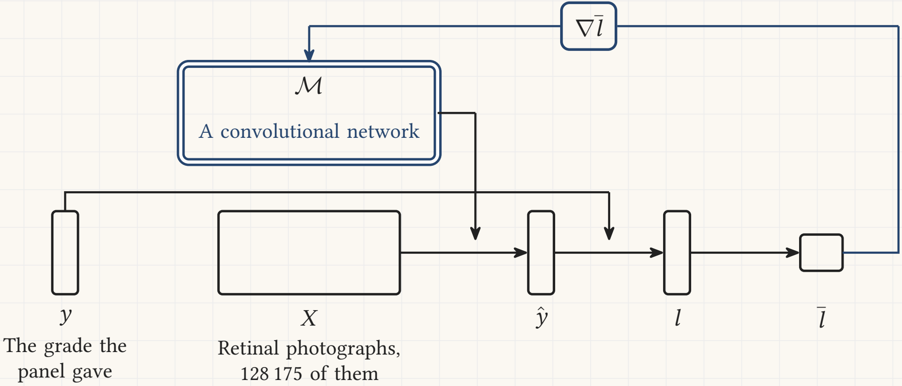

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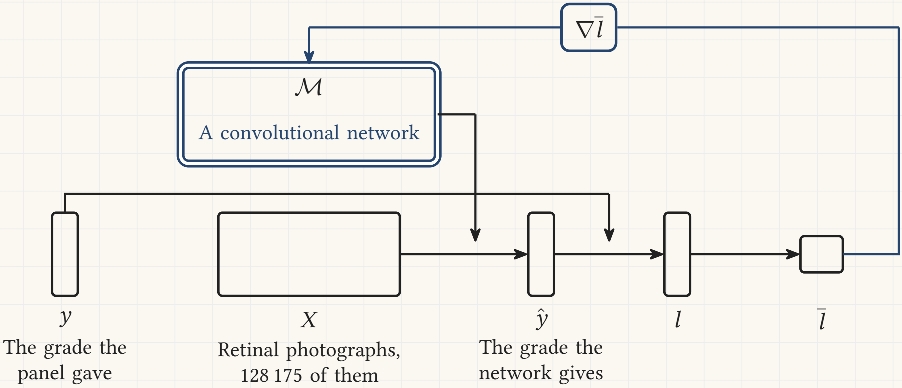

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